

D864

Secondary Physics Curriculum

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

Method © 2026 Erik Daniel Lindquist — Course Mastery Coach, armored drill. Items drafted with AI assistance under his direction.

How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word _____	The one fact that decides _____	Answer _____	[] sure [] unsure [] guess
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Items

1.

A graduate assistant in a well-funded materials lab, preparing a conference poster, finds that 2 of 20 trial runs do not fit the predicted curve. The poster is due Monday, and the lab's coffee machine just broke. The assistant considers adjusting those two values slightly so the trend looks cleaner. What would this be?

- A. Plagiarism.
- B. Falsification; the assistant should report all 20 runs and investigate the two outliers.
- C. Acceptable, since only 10% of the data would change.
- D. Acceptable as long as the final conclusion is correct.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2.

During a Praxis study group held in a library meeting room, a teacher candidate says, "Once the theory of gravity is proven enough, it becomes a law." The group meets every Tuesday. Which correction is accurate?

- A. A theory is just an educated guess.
- B. The candidate is right; theories become laws once proven.
- C. Laws describe what happens; theories explain why. A theory does not turn into a law.
- D. Laws are never revised once stated.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

A roller-coaster car at an amusement park that opened in 1972 starts from rest at the top of a 20 m hill. The car is painted red and holds 12 riders. Ignoring friction, what is its speed at the bottom? ($g = 9.8 \text{ m/s}^2$)

- A. It depends on the riders' total mass.
- B. About 19.8 m/s
- C. 392 m/s
- D. 14 m/s

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4. [TRANSFER — reversed stem / different variable]

A 2.0 kg ball is dropped and hits the ground at 10 m/s. Ignoring air resistance, from what height was it dropped? ($g = 9.8 \text{ m/s}^2$)

- A.** About 0.51 m
- B.** It depends on the 2.0 kg mass.
- C.** About 5.1 m
- D.** About 10.2 m

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

A teacher, whose lab has a new refrigerator for storing samples, asks students why a refrigerator needs electric power to move heat out of its cold interior into the warmer room. The fridge has an efficiency rating label. Which explanation is best?

- A.** A refrigerator creates cold, which takes energy.
- B.** Heat naturally flows from cold to hot; the fridge just speeds it up.
- C.** Heat flows on its own only from hot to cold; moving it from cold to hot requires work.
- D.** An efficient fridge needs no power to move heat.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

Two charged dust particles in an air-filter demo, designed by an engineering club, attract with a force of $18 \mu\text{N}$ when 1 cm apart. The filter housing is clear plastic. What is the force when they are 3 cm apart?

- A.** $18 \mu\text{N}$, since the charges did not change.
- B.** $6 \mu\text{N}$
- C.** $162 \mu\text{N}$
- D.** $2 \mu\text{N}$

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7. [TRANSFER — reversed stem / different variable]

Two charges attract with force F . What change in their separation would make the force $4F$?

- A.** Halve the separation.
- B.** Separation does not matter; only charge does.
- C.** Reduce the separation to one-quarter.
- D.** Double the separation.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

A string of holiday lights, sold by a discount store, has its bulbs wired in series across 120 V. One bulb burns out. The string is 8 m long. What happens?

- A. All the bulbs go out, because the break opens the only path for current.
- B. Only the burned-out bulb goes dark.
- C. The other bulbs get brighter.
- D. The string keeps working, because 120 V is enough to jump the gap.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9. [TRANSFER — reversed stem / different variable]

Resistors of 2 Ω , 4 Ω , and 6 Ω are connected in series with a 24 V battery. What current flows?

- A. About 22 A
- B. 12 A
- C. 0.5 A
- D. 2 A

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

A teacher, designing a wall poster with graphics from a public space-agency website, orders the electromagnetic spectrum. Which list runs from lowest to highest photon energy?

- A. Gamma, X-ray, ultraviolet, visible, infrared, microwave, radio.
- B. All bands carry the same photon energy, since they all travel at c.
- C. Radio, infrared, microwave, visible, ultraviolet, X-ray, gamma.
- D. Radio, microwave, infrared, visible, ultraviolet, X-ray, gamma.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11. [TRANSFER — reversed stem / different variable]

Compared with visible light, X-rays in a vacuum have:

- A. The same speed, a higher frequency, and a shorter wavelength.
- B. A lower frequency and a longer wavelength.
- C. A higher frequency and a longer wavelength.
- D. A higher speed.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

In a classic demo with a bell jar bought decades ago, a ringing electric bell is placed inside and the air is pumped out. The clapper is still visibly striking the bell. The jar is 30 cm tall. What do students observe, and why?

- A.** The sound gets louder as air is removed.
- B.** The sound fades away, because sound needs a medium, while light still carries the image of the bell.
- C.** The sound stays the same, because sound travels through a vacuum like light.
- D.** The bell stops ringing in a vacuum.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	____	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer B Concept 1

A graduate assistant in a ~~well-funded~~ materials lab, preparing a conference poster, finds that 2 of 20 trial runs do not fit the predicted curve. ~~The poster is due Monday, and the lab's coffee machine just broke.~~ The assistant considers adjusting those two values slightly so the trend looks cleaner. What would this be?

Target word: classify the action **Decides:** Changing recorded data.

A	Confusion set	Plagiarism is using others' work without credit; this is altering data.
B	Correct	Altering data to fit a prediction is research misconduct.
C	Wrong stem number	The share of data changed does not make falsification acceptable.
D	Adjacent category	A true conclusion does not excuse altered evidence.

2. Answer C Concept 2

During a Praxis study group ~~held in a library meeting room~~, a teacher candidate says, "Once the theory of gravity is proven enough, it becomes a law." ~~The group meets every Tuesday.~~ Which correction is accurate?

Target word: theory vs. law **Decides:** Laws describe; theories explain.

A	Adjacent category	That describes a hypothesis; a theory is a well-tested explanation.
B	Inverted operation	Reverses the relationship; the two do different jobs.
C	Correct	They are different kinds of knowledge, not steps on one ladder.
D	Absolute language	Laws can be refined when new evidence appears.

3. Answer B Concept 3

A roller-coaster car ~~at an amusement park that opened in 1972~~ starts from rest at the top of a 20 m hill. ~~The car is painted red and holds 12 riders.~~ Ignoring friction, what is its speed at the bottom? ($g = 9.8 \text{ m/s}^2$)

Target word: speed at the bottom **Decides:** $v = \sqrt{2gh}$.

A	Camouflage	Mass cancels; speed is independent of it.
B	Correct	$\sqrt{2 \times 9.8 \times 20} = \sqrt{392} \approx 19.8 \text{ m/s}$.
C	Dropped element	Drops the square root.
D	Confusion set	Uses $mgh = mv^2$, forgetting the $\frac{1}{2}$ in kinetic energy.

4. Answer C Concept 3 • transfer

A 2.0 kg ball is dropped and hits the ground at 10 m/s. Ignoring air resistance, from what height was it dropped? ($g = 9.8 \text{ m/s}^2$)

Target word: height from impact speed **Decides:** $h = v^2/(2g)$.

A	Confusion set	Uses v instead of v^2 .
B	Adjacent category	Mass cancels in energy conservation.
C	Correct	$100 \div 19.6 \approx 5.1 \text{ m}$.
D	Dropped element	Drops the 2 in $2g$.

5. Answer C Concept 4

A teacher, whose lab has a new refrigerator ~~for storing samples~~, asks students why a refrigerator needs electric power to move heat out of its cold interior into the warmer room. ~~The fridge has an efficiency rating label.~~ Which explanation is best?

Target word: why work is needed **Decides:** Heat moved from cold to hot.

A	Confusion set	"Cold" is not made; heat is removed.
B	Inverted operation	Reverses the natural direction of heat flow.
C	Correct	This is the second law of thermodynamics.
D	Absolute language	Efficiency lowers the work needed but cannot remove it.

6. Answer D Concept 5

Two charged dust particles in an air-filter demo, ~~designed by an engineering club~~, attract with a force of 18 μN when 1 cm apart. ~~The filter housing is clear plastic.~~ What is the force when they are 3 cm apart?

Target word: force at triple distance **Decides:** $F \propto 1/r^2$.

A	Adjacent category	Distance matters as much as charge.
B	Dropped element	Divides by 3, dropping the square.
C	Inverted operation	Multiplies by 9; force falls with distance.
D	Correct	$18 \div 3^2 = 18 \div 9 = 2 \mu\text{N}$.

7. Answer A Concept 5 • transfer

Two charges attract with force F. What change in their separation would make the force 4F?

Target word: separation for a given force **Decides:** $4 = (1/k)^2 \rightarrow k = \frac{1}{2}$.

A	Correct	$(1/0.5)^2 = 4$.
B	Absolute language	Force depends strongly on distance.
C	Dropped element	Drops the square; that would give 16F.
D	Inverted operation	Moving apart weakens the force.

8. Answer A Concept 6

A string of holiday lights, ~~sold by a discount store~~, has its bulbs wired in series across 120 V. One bulb burns out. ~~The string is 8 m long.~~ What happens?

Target word: effect of an open bulb **Decides:** Single path.

A	Correct	Series circuits have one loop.
B	Confusion set	That is how parallel wiring behaves.
C	Adjacent category	Brightening happens when a bulb shorts, not when it opens.
D	Camouflage	The voltage rating does not bridge an open filament.

9. Answer D Concept 6 · transfer

Resistors of $2\ \Omega$, $4\ \Omega$, and $6\ \Omega$ are connected in series with a $24\ \text{V}$ battery. What current flows?

Target word: current in a series circuit **Decides:** $R_{\text{total}} = 12\ \Omega$.

A	Confusion set	Combines the resistors as if they were in parallel.
B	Wrong stem number	Uses only the $2\ \Omega$ resistor.
C	Inverted operation	Divides R by V .
D	Correct	$24 \div 12 = 2\ \text{A}$.

10. Answer D Concept 7

A teacher, designing a wall poster ~~with graphics from a public space agency website~~, orders the electromagnetic spectrum. Which list runs from lowest to highest photon energy?

Target word: order by energy **Decides:** Energy rises with frequency.

A	Inverted operation	That runs from highest to lowest.
B	Absolute language	Same speed, very different energies.
C	Confusion set	Swaps microwave and infrared.
D	Correct	Frequency and photon energy increase in that order.

11. Answer A Concept 7 · transfer

Compared with visible light, X-rays in a vacuum have:

Target word: compare X-rays with visible light **Decides:** Higher frequency, same speed.

A	Correct	All EM waves travel at c ; X-rays are higher in frequency.
B	Inverted operation	Reverses the X-ray position on the spectrum.
C	Adjacent category	At fixed speed, higher frequency means shorter wavelength.
D	Confusion set	All EM waves share the same vacuum speed.

12. Answer B Concept 8

In a classic demo with a bell jar ~~bought decades ago~~, a ringing electric bell is placed inside and the air is pumped out. The clapper is still visibly striking the bell. ~~The jar is 30 cm tall.~~ What do students observe, and why?

Target word: observation and cause **Decides:** Air removed; light still passes.

A	Inverted operation	Removing the medium weakens the sound.
B	Correct	Sound is a mechanical wave.
C	Confusion set	Mixes up mechanical and electromagnetic waves.
D	Adjacent category	The clapper is still striking; the bell rings but the sound cannot reach the students.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	4	
Wrong stem number	A figure copied from the prompt that is not the answer.	2	
Adjacent category	The right idea in the wrong bucket.	7	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	8	
Absolute language	Always, never, guarantee — where judgment applies.	4	
Confusion set	Every option is real; the test is exact role-matching.	9	
Camouflage	Answers a stem fact that was never the question.	2	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Research misconduct	Fabrication (making up data), falsification (changing or omitting data), and plagiarism are misconduct; anomalous data are reported and investigated, not adjusted.	Treating a small tweak to “clean up” data as acceptable.	A researcher tempted to change a few values.
2	Theory vs. law	A law describes what happens (often as an equation); a theory explains why. Theories do not become laws when “proven.”	Seeing law as a promoted, more certain theory.	A claim that a theory will “become a law.”
3	Conservation of mechanical energy	Without friction, potential energy lost equals kinetic energy gained: $mgh = \frac{1}{2}mv^2$, so $v = \sqrt{2gh}$, independent of mass.	Forgetting the square root or the $\frac{1}{2}$, or thinking heavier objects go faster.	A height and a speed with friction ignored.
4	Second law of thermodynamics	Heat flows on its own only from hot to cold; moving heat from cold to hot requires work, and no engine turns all heat into work.	Thinking a refrigerator “makes cold.”	Heat moved against the natural direction.
5	Coulomb’s law and distance	Electric force varies as $1/r^2$: tripling distance cuts force to one-ninth; halving distance quadruples it.	Using $1/r$ instead of $1/r^2$.	A force at one separation and a question at another.
6	Series circuits	In series, one path carries the same current through every part; resistances add; a break anywhere stops all current.	Treating a series string like parallel branches.	Components on a single loop.
7	The electromagnetic spectrum	From radio to gamma rays, frequency and photon energy rise while wavelength falls; all travel at c in a vacuum.	Thinking higher-energy waves travel faster, or reversing the order.	A list of EM bands to order or compare.
8	Sound needs a medium	Sound is a mechanical wave and cannot travel through a vacuum; light can.	Assuming sound travels like light.	A sound source in a jar being emptied of air.

Course scope (public description): Secondary Physics Curriculum. This course offers a comprehensive review for high school physics teachers preparing for the Praxis exam. Explore major scientific discoveries, key engineering innovations, and their societal impacts. Understand ethical issues in scientific research and the principles of effective science communication. Delve into different forms of energy, laws of thermodynamics, and the role of energy and matter in the universe. Study Coulomb's Law, electric circuits, and electromagnetism. Learn how to integrate wave properties, sound waves, and the electromagnetic spectrum into your teaching practice. As this course is focused on Praxis prep, content is conceptual and algebra- based.